

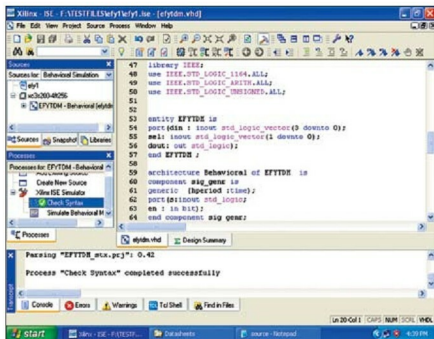


Fig. 5: Result of run command output

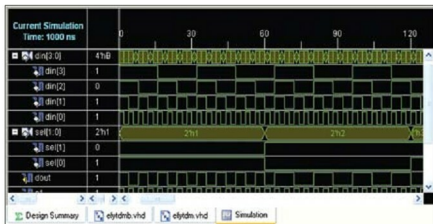


Fig. 6: EFTDM signal with multiplexed wave on 'dout' signal line

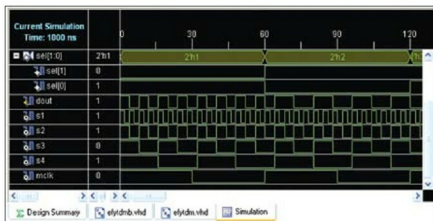


Fig. 7: EFTDM output with internal signals s1, s2, s3 and s4

The master clock 'mclk' is also generated from the component 'sig_gen' with the half period of 30 ns.

```
mstrclk: sig_gen generic map (30ns)
port map (mclk, '1');
```

The 'sel' select signals are generated from the master clock 'mclk' using a concept of the 'divide by' counter. Here, a 2-bit (divide by 2) counter is used. The process 'sel' in the TDM architecture is a code that generates four select signals from the master clock 'mclk' by the 2-bit counter (divide by 2) concept (Fig. 1). The outputs of the 2-bit counter are the select signals that will be used in multiplexing. The function CONV_STD_LOGIC_VECTOR (defined in the IEEE library) converts an integer value into std_logic_vector. The remaining code that is used is self explanatory.

```
sel: process (mclk)
variable temp:integer:=0;
begin
if mclk = '1' and mclk'event then
temp := temp +1;
end if;
sel<= CONV_STD_LOGIC_VECTOR(temp,2);
end process sel;
```

Testing steps

You need to install the latest version of Integrated Software Environment (ISE) Xilinx 9.2i in your system. The test bench and test bench waveform used in this simulation are under the Xilinx ISE platform. The test bench is used for ensuring that the design works properly. To correctly test the system, you need to select the following options before executing the code:

1. Open a new project from file menu option
2. Enter the project name as, say, EFTY1
3. Leave this step as the default if everything is the same (Fig. 3); otherwise select the relevant options by clicking the right arrow marks followed by the 'Next' button without loading the source code.
4. In the 'Processes' window, click the 'Create new source' option and the 'Execute' option. Select the VHDL